

Superconductivity in 2D Multicomponents Materials

Time-reversal symmetry breaking in 2D multi-components material with Josephson coupling in the London limit using Metropolis algorithm in the Monte Carlo simulation

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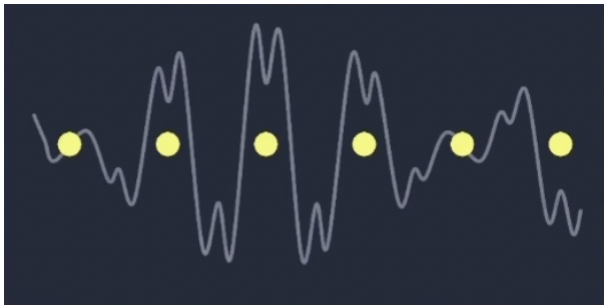
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Theory in superconductivity

- Ohm's law : $R = \frac{U}{I}$
- Superconductivity (1911) $\rightarrow$ no resistance.
- Duality wave-particle in metals by Bloch (1930):

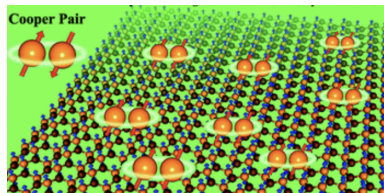
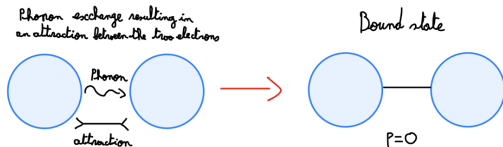


Cooper pairs

- Order Parameter:

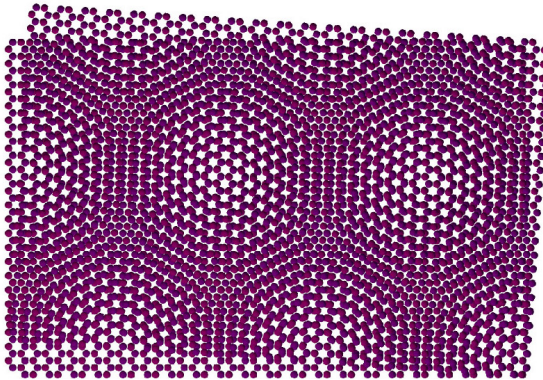
$$O_s = \int \frac{\Psi}{A} dx$$

Cooper pair formation



Moiré materials

- Twisting of two 2D graphene from one to another:



Josephson coupling in the London limit

- Ginzburg-Landau free energy:

$$f = \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4 + \frac{1}{2m}|(-i\hbar\nabla - 2e\vec{A})\Psi|^2 + \frac{\mu_0 B^2}{2}$$

- But in 2D:

$$f = \alpha|\Psi|^2 + \frac{\beta}{2}|\Psi|^4 + \frac{1}{2m}|\nabla\Psi|^2$$

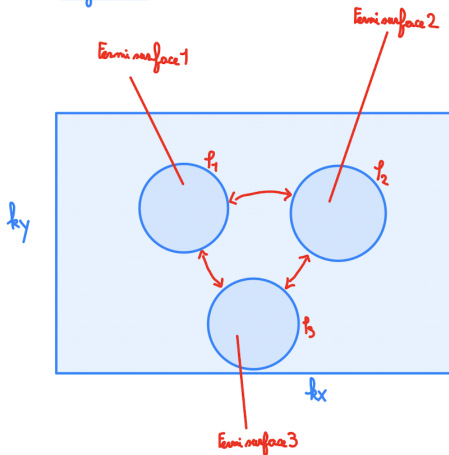
- In our scenario:

$$f = \alpha_i|\Psi_i|^2 + \frac{\beta_i}{2}|\Psi_i|^4 + \frac{1}{2m}|\nabla\Psi_i|^2 + \epsilon_{ij}\Psi_i\Psi_j^* + \text{constant}$$



Josephson coupling in the London limit

System:



Each superconductivity component forms on a Fermi surface. p_i belongs to Fermi surface i .

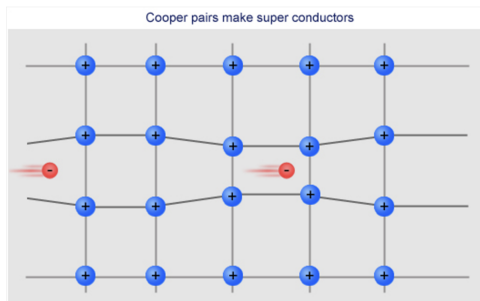
Time-reversal symmetry

- Time reversal operator T : $T\Psi(x, t) = \Psi^*(x, -t)$
- $T = UK$, U = unitary operator, K = complex conjugate operator
- Kramer's theorem: $T^2 = \pm 1$
- $TpT^{-1} = -p$ (=anti-commute)
- $TJT^{-1} = -J$
- $TxT^{-1} = x$ (=commute)
- GL-theory $\rightarrow$ non vanishing order parameter $S \rightarrow$ broken time-reversal symmetry $TS = -S$



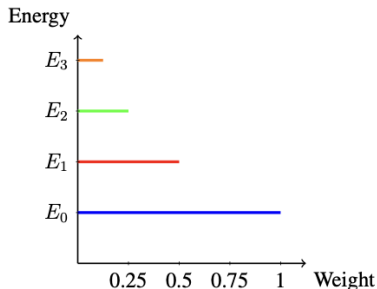
Phase stiffness(2D), $U(1)$ symmetry(3D)

- Phase stiffness $\rightarrow$ energy required to change the phase order $\rightarrow$ lattice distortion
- $U(1)$ symmetry $\rightarrow$ gauge symmetry conserving laws of physics under transformations to fields and particle
- Cooper pairs $\rightarrow U(1)$ symmetry spontaneously broken.



- Boltzmann factor:

$$P(E_i) = \sum_i e^{-\beta H_i}; \beta = \frac{1}{kT}$$



- Hamiltonian:

$$H = - \sum_{A, \langle i, j \rangle} \cos(\varphi_{A,i} - \varphi_{A,j}) + \eta \sum_{i, A \neq B} \cos(\varphi_{A,i} - \varphi_{B,i})$$



Monte Carlo Simulation

- (1)-Input values
- (2)-New example sample of values
- (3)-Mathematical model
- (4)-Output the accepted values
- (5)-Analysis of the data given



Metropolis Algorithm

- Generating/ Selecting an initial value θ_0
- The number of simulations = $n \rightarrow$ we repeat n times:
 - New value proposed θ^* from proposal probability distribution $p(\theta^*|\theta_n)$
 - Rate: $r = \frac{p(\theta^*)p(\theta_n|\theta^*)}{p(\theta_n)p(\theta^*|\theta_n)}$
 - If $r > 1$: $\theta_n = \theta^*$
 - If $1 > r > 0$: $\theta_n = \theta^*$ is r probable and $\theta_n = \theta_0$ is $1 - r$ probable.
 - $\theta_n =$ new initial values $\rightarrow$ compared with a new proposed value in next simulation.
- Probability density symmetrical $\rightarrow \frac{p(\theta_n|\theta^*)}{p(\theta^*|\theta_n)} = 1 \rightarrow r = \frac{p(\theta^*)}{p(\theta_n)}$



The code

- Boltzmann factor/ Probability of the Hamiltonian($k = 1$):

$$\text{prob}(H) = e^{-\frac{H}{T}}$$

- Ratio:

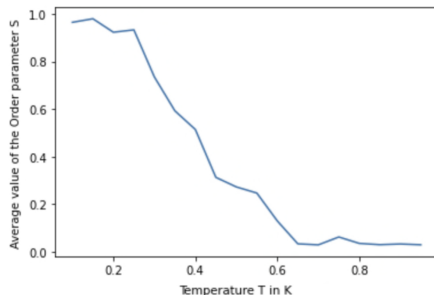
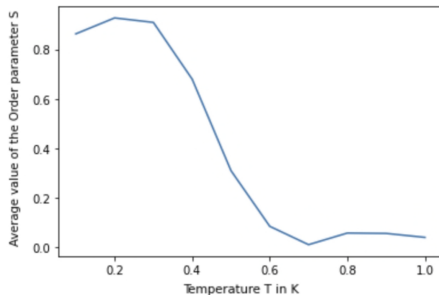
$$r = \frac{\text{prob}(H_2)}{\text{prob}(H_1)}$$

- if p_1, p_2, p_3 in one order $\rightarrow$ phase order "up".
- Else, phase order "down".

Example of phase order "up" and "down"

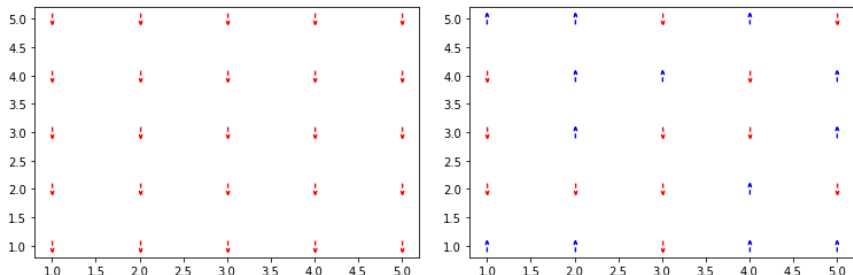


- Average order parameter $\langle |S| \rangle$ related to time-reversal symmetry in function of the temperature:



Results

- Order parameter S_i related to time-reversal symmetry for a lattice 5x5 at $T=0.5\text{K}$ and $T=10\text{K}$:



Flaws, error and accuracy

- Standard deviation

$$\sigma = \sqrt{\frac{\langle |S|^2 \rangle - \langle |S| \rangle^2}{N}}$$

- with

$$\langle |S| \rangle = \frac{1}{N} \sum_{n=1}^N |S_n|$$

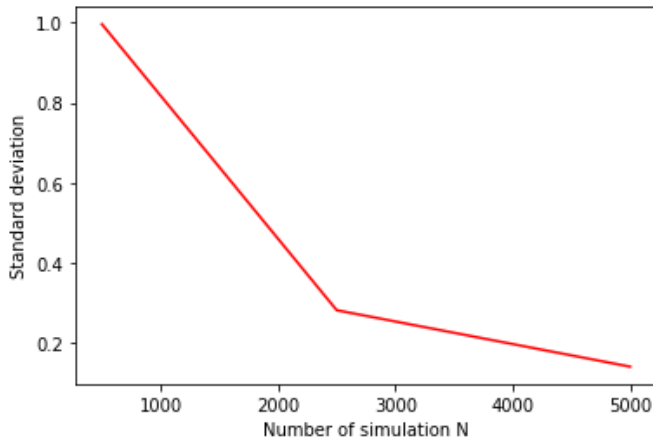
and

$$\langle |S|^2 \rangle = \frac{1}{N} \sum_{n=1}^N |S_n|^2$$



Flaws, error and accuracy

- Standard deviation over the total number of simulation N :



Flaws, error and accuracy

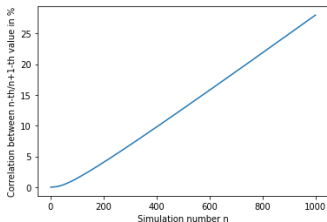
- Correlation function between the simulations:

$$c(l) = \sqrt{\frac{1}{N-l} \sum_{n=1}^{N-l} (S_n - \langle S \rangle)(S_{n+l} - \langle S \rangle)}$$

- When $\langle S \rangle = 0$,

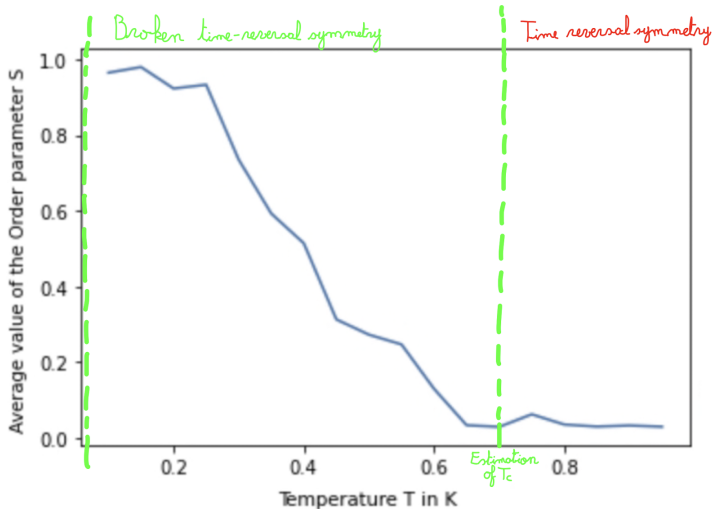
$$c(l) = \sqrt{\frac{1}{N-l} \sum_{n=1}^{N-l} S_n S_{n+l}}$$

- Correlation between n -th and $n+1$ -th value over the simulation number n :



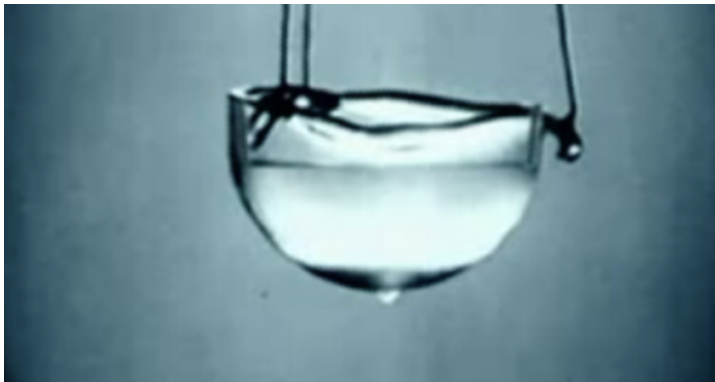
Interpretation

- Separation of the normal time-reversal symmetry and spontaneously broken time-reversal symmetry state:



Conclusion

- Picture of a superfluid liquid helium:



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